YeshwantraoChavan College of Engineering
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Department of Applied Mathematics & Humanities

Electronic Question Answer Bank

Course Name: Calculus and Vector **Course Code: 23GE1101**

Unit 5 :- Vector Calculus

1	Unit 5	Given the vector field Find the curl . Sol^n: $\bar{v} = (x^2-y^2+2xz)i + (xz-xy+yz)j + (z^2+x^2)k$. $\therefore \text{curl}\,\bar{v} = \nabla \times \bar{v}$ $= \begin{vmatrix} i & j & k \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ x^2-y^2+2xz & xz-xy+yz & z^2+x^2 \end{vmatrix}$ $= i[0-(x+y)] - j[2x-2x] + k[z-y+2y]$ $\therefore \text{curl}\,\bar{v} = -(x+y)i - 0j + (z+y)k$.	10	5	Low	Easy	CO-4	Numerical

2	Unit 5	Find the directional derivatives of in the direction of r, where	10	5	Low	Easy	CO-4	Numerical		

Soln. Let $\phi = \frac{1}{r} = r^{-1}$ $\quad [\because \bar{r} = xi + yj + zk]$

Directional derivative $= \bar{a} \cdot \text{grad}\ \phi$

$$\bar{a} = \frac{x\,i + y\,j + z\,k}{\sqrt{x^2 + y^2 + z^2}}$$

Now, $r^2 = x^2 + y^2 + z^2$ ..... (A)

$$\therefore r = \sqrt{x^2 + y^2 + z^2}$$

$$\bar{a} = \frac{x}{r}i + \frac{y}{r}j + \frac{z}{r}k$$

Now,

$$\nabla\phi = \frac{\partial\phi}{\partial x}i + \frac{\partial\phi}{\partial y}j + \frac{\partial\phi}{\partial z}k$$

$$\frac{\partial\phi}{\partial x} = \frac{\partial}{\partial x}r^{-1} = -r^{-2}\frac{\partial r}{\partial x} \quad \ldots\ldots(2)$$

Differentiating (A) w.r.t. 'x'

$$2r\frac{\partial r}{\partial x} = 2x \Rightarrow \frac{\partial r}{\partial x} = \frac{x}{r}$$

From equation (2) $\Rightarrow$ $\frac{\partial\phi}{\partial x} = -r^{-2}\frac{x}{r}$

$$\therefore \frac{\partial\phi}{\partial x} = -\frac{x}{r^3}$$

Similarly, $\frac{\partial\phi}{\partial y} = \frac{-y}{r^3}$ and $\frac{\partial\phi}{\partial z} = \frac{-z}{r^3}$

From equation (1) $\Rightarrow$

$$\nabla\phi = \frac{-x}{r^3}i - \frac{y}{r^3}j - \frac{z}{r^3}k$$

$$\therefore \nabla\phi = \frac{-1}{r^3}(xi + yj + zk)$$

Directional derivative $= \bar{a}$. grad ϕ

$$= \left(\frac{x}{r}i + \frac{y}{r}j + \frac{z}{r}k\right) \cdot \left(\frac{-1}{r^3}(xi + yj + zk)\right)$$

$$= -\frac{1}{r^4}\left(x^2 + y^2 + z^2\right)$$

$$= -\frac{1}{r^4}.r^2 \qquad [\because \text{By (A)}]$$

$$\therefore \boxed{\text{Directional derivative} = -\frac{1}{r^2}} \quad \textbf{Ans.}$$

3	Unit 5	A vector field $\overline{A}$ is given by prove that it is irrotational and hence find its scalar potential. **Soln.** We have, $\overline{F} = (y\sin z - \sin x)\,i + (x\sin z + 2yz)\,j + (xy\cos z + y^2)\,k$ $\Rightarrow \nabla \times \overline{F} = \begin{vmatrix} i & j & k \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ y\sin z - \sin x & x\sin z + 2yz & xy\cos z + y^2 \end{vmatrix}$ $= i\left\{\frac{\partial}{\partial y}(xy\cos z + y^2) - \frac{\partial}{\partial z}(x\sin z + 2yz)\right\}$ $- j\left\{\frac{\partial}{\partial x}(xy\cos z + y^2) - \frac{\partial}{\partial z}(y\sin z - \sin x)\right\}$ $+ k\left\{\frac{\partial}{\partial x}(x\sin z + 2yz) - \frac{\partial}{\partial y}(y\sin z - \sin x)\right\}$ $\therefore \nabla \times \overline{F} = i(x\cos z + 2y) - (x\cos z + 2y)]$ $-j(y\cos z - y\cos z) + k(\sin z - \sin z)]$ $= 0$ Hence $\overline{F}$ is irrotational. **Ans.** If ϕ is the scalar potential, then $\overline{F} = \nabla\phi$ $\Rightarrow (y\sin z - \sin x)\,i + (x\sin z + 2yz)\,j + (xy\cos z + y^2)k$ $= \frac{\partial\phi}{\partial x}i + \frac{\partial\phi}{\partial y}j + \frac{\partial\phi}{\partial z}k$ $\Rightarrow \frac{\partial\phi}{\partial x} = y\sin z - \sin x$ ..... (1) $\frac{\partial\phi}{\partial y} = x\sin z + 2yz$ ..... (2) $\frac{\partial\phi}{\partial z} = xy\cos z + y^2$ ..... (3) Integrating equations (1), (2), (3) partially w. r. to x, y, z respectively, we get $\phi = xy\sin z + \cos x + \phi_1(y, z)$ ..... (4) $\phi = xy\sin z + y^2 z + \phi_2(x, z)$ ..... (5) $\phi = xy\sin z + y^2 z + \phi_3(x, y)$ ..... (6) Comparing equations (4), (5) & (6), we get $\phi_1(y,z)$ = terms in ϕ independent of $x = y^2 z$ $\phi_2(x,z)$ = terms in ϕ independent of $y = \cos x$ $\phi_3(x,y)$ = terms in ϕ independent of $z = \cos x$ $\phi = xy\sin z + \cos x + y^2 z$ $\therefore$ $\boxed{\phi = xy\sin z + \cos x + y^2 z}$ **Ans.**	10	5	Low	Easy	CO-4	Numerical
4	Unit 5	Find the directional derivatives of in the direction of , where	10	5	Low	Easy	CO-4	Numerical

Soln. Let $\phi = \frac{1}{r} = r^{-1}$ $\quad [\because \ \bar{r} = xi + yj + zk]$

Directional derivative $= \bar{a}$. grad ϕ

$$\bar{a} = \frac{x\,i + y\,j + z\,k}{\sqrt{x^2 + y^2 + z^2}}$$

Now, $r^2 = x^2 + y^2 + z^2$ ..... (A)

$$\therefore r = \sqrt{x^2 + y^2 + z^2}$$

$$\bar{a} = \frac{x}{r}i + \frac{y}{r}j + \frac{z}{r}k$$

Now,

		$\nabla\phi = \frac{\partial\phi}{\partial x}i + \frac{\partial\phi}{\partial y}j + \frac{\partial\phi}{\partial z}k$ $\frac{\partial\phi}{\partial x} = \frac{\partial}{\partial x}r^{-1} = -r^{-2}\frac{\partial r}{\partial x}$ ..... (1) Differentiating (A) w.r.t. 'x' $2r\frac{\partial r}{\partial x} = 2x \Rightarrow \frac{\partial r}{\partial x} = \frac{x}{r}$ ..... (2) From equation (2) $\Rightarrow$ $\frac{\partial\phi}{\partial x} = -r^{-2}\frac{x}{r}$ $\therefore \frac{\partial\phi}{\partial x} = -\frac{x}{r^3}$ Similarly, $\frac{\partial\phi}{\partial y} = \frac{-y}{r^3}$ and $\frac{\partial\phi}{\partial z} = \frac{-z}{r^3}$ From equation (1) $\Rightarrow$ $\nabla\phi = \frac{-x}{r^3}i - \frac{y}{r^3}j - \frac{z}{r^3}k$ $\therefore \nabla\phi = \frac{-1}{r^3}(xi + yj + zk)$ Directional derivative $= \bar{a}$. grad ϕ $= \left(\frac{x}{r}i + \frac{y}{r}j + \frac{z}{r}k\right) \cdot \left(\frac{-1}{r^3}(xi + yj + zk)\right)$ $= -\frac{1}{r^4}\left(x^2 + y^2 + z^2\right)$ $= -\frac{1}{r^4}.r^2$ *[$\because$ By (A)]* $\therefore$ **Directional derivative $= -\frac{1}{r^2}$** **Ans.**						
5	Unit 5	If find and unit normal at the point .	10	5	Low	Easy	CO-4	Numerical

		$\phi = 3x^2y - y^3z^2$ $\frac{\partial\phi}{\partial x} = 6xy \quad \frac{\partial\phi}{\partial y} = 3x^2 - 3y^2z^2 \quad \frac{\partial\phi}{\partial z} = -2y^3z$ $\therefore \text{grad}\,\phi = \nabla\phi$ $= i\frac{\partial\phi}{\partial x} + j\frac{\partial\phi}{\partial y} + k\frac{\partial\phi}{\partial z}$ $= 6xyi + (3x^2 - 3y^2z^2)j - 2y^3z\,k$ $(\text{grad}\,\phi)_{(1,-2,-1)} = -12i - 9j + 16k$ unit normal vector $\hat{n} = \frac{\nabla\phi}{\lvert\nabla\phi\rvert}$ $= \frac{-12i - 9j + 16k}{\sqrt{144 + 81 + 256}}$ $\hat{n} = \frac{-12i - 9j + 16k}{\sqrt{481}}$						
6	Unit 5	If . Then find $u = x^2 + y^2 + z^2, \quad \bar{r} = xi + yj + zk$ $\nabla u = 2xi + 2yj + 2zk \quad \nabla\bar{r} = 3$ $\therefore \text{div}(u\bar{r}) = \nabla\cdot(u\bar{r})$ $= u.(\nabla\bar{r}) + (\nabla u)\bar{r}$ $= 3u + (2xi + 2yj + 2zk)(xi + yj + zk)$ $= 3u + 2(x^2 + y^2 + z^2)$ $= 3u + 2u$ $= 5u.$	10	5	Low	Easy	CO-4	Numerical
7	Unit 5	At any point of the curve find (i) tangent vector (ii) unit tangent vector.	10	5	Low	Easy	CO-4	Numerical

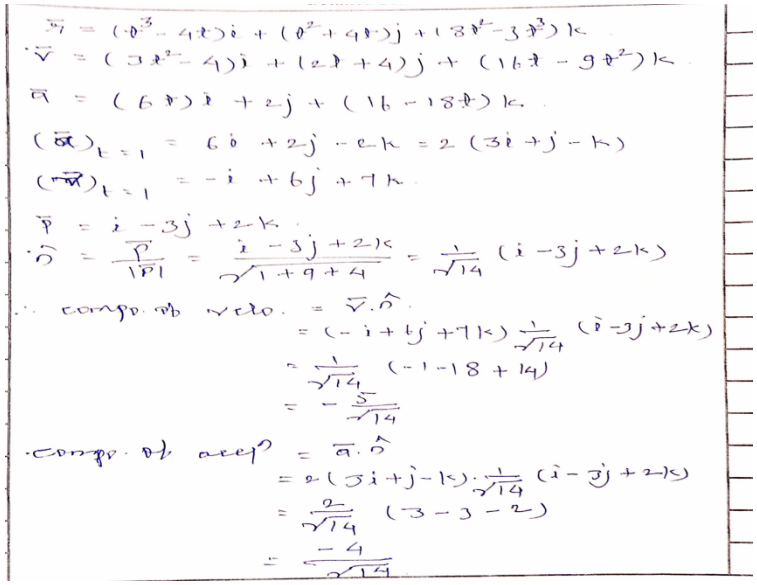
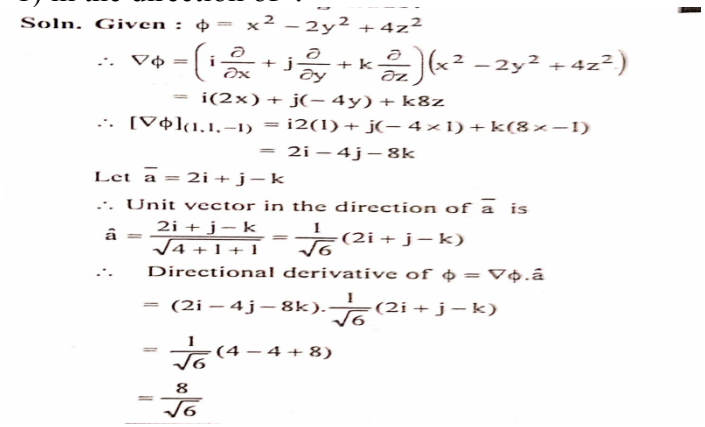
		$x = 3\cos t \quad y = 3\sin t \quad z = 4t$ $\bar{r} = xi + yj + zk$ $\bar{r} = 3\cos t\, i + 3\sin t\, j + 4t\, k.$ (i) tangent vector $= \frac{d\bar{r}}{dt}$ $= -3\sin t\, i + 3\cos t\, j + 2t^2 k.$ (ii) unit tangent vector $= \frac{d\bar{r}/dt}{\lvert d\bar{r}/dt \rvert}$ $= \frac{-3\sin t\, i + 3\cos t\, j + 2t^2 k}{\sqrt{9\sin^2 t + 9\cos^2 t + 4t^4}}$ $= \frac{-3\sin t\, i + 3\cos t\, j + 2t^2 k}{\sqrt{9 + 4t^4}}$							
8	Unit 5	Show that div, where. $\operatorname{div}\left(\frac{\bar{r}}{r^3}\right) = \nabla\left(r^{-3}\cdot\bar{r}\right)$ $= r^{-3}(\nabla\bar{r}) + (\nabla r^{-3})\bar{r}$ $= 3r^{-3} + (-3)r^{-5}\,\bar{r}\cdot\bar{r}$ $= \frac{3}{r^3} - \frac{3}{r^5}\cdot r^2$ $= \frac{3}{r^3} - \frac{3}{r^3}$ $= 0.$	10	5	Low	Easy	CO-4	Numerical

09	Unit 5	If , find (i) (ii) (iii) so that .	12	5	Medium	Moderate	CO-4	Numerical
10	Unit 5	A particle moves along the curve wheret is the time. Find the magnitude of tangential components of its acceleration at . **Soln. Given:** $\bar{r} = (t^3 - 4t)\,\bar{i} + (t^2 + 4t)\,\bar{j} + (8t^2 - 3t^3)\,\bar{k}$ Velocity, $\bar{v} = \frac{d\bar{r}}{dt} = (3t^2 - 4)\,\bar{i} + (2t + 4)\,\bar{j} + (16t - 9t^2)\,\bar{k}$ Acceleration, $\bar{a} = \frac{d^2\bar{r}}{dt^2} = 6t\,\bar{i} + 2\bar{j} + (16 - 18t)\,\bar{k}$ At $t = 2$, $\bar{v} = 8i + 8j - 4k$, $\bar{a} = 12i + 2j - 20k$ Unit tangent vector $\hat{t} = \frac{\bar{v}}{\lvert\bar{v}\rvert} = \frac{8i + 8j - 4k}{\sqrt{64 + 64 + 16}} = \frac{8i + 8j - 4k}{12}$ Tangential component of acceleration, $a_t = \bar{a} \cdot \hat{t}$ $= (12i + 2j - 20k) \cdot \frac{8i + 8j - 4k}{12}$ $= \frac{12 \times 8 + 2 \times 8 + 20 \times 4}{12}$ $= \frac{192}{12}$ $= 16$	12	5	Medium	Moderate	CO-4	Numerical
11	Unit 5	Find the directional derivative of the function at the point (1,1,-1) in the direction towards the point (-3,5,6).	12	5	Medium	Moderate	CO-4	Numerical

		Soln. Given : $\phi = 4e^{2x-y+z}$ The directional derivative of ϕ at (1, 1, – 1) in the direction towards the point $(-3,5,6)$ is given by $\frac{\partial\phi}{\partial r} = [\nabla\phi_{(1,1,-1)}] \cdot \left[\begin{matrix}\text{Unit vector in the direction}\\ \text{towards the point}(-3,5,6)\end{matrix}\right]$ ....(1) Now, $\nabla\phi = \frac{\partial\phi}{\partial x}i + \frac{\partial\phi}{\partial y}j + \frac{\partial\phi}{\partial z}k$ $= (8e^{2x-y+z})i + (-4e^{2x-y+z})j + (4e^{2x-y+z})k$ $\therefore (\nabla\phi)_{(1,1,-1)} = 8i - 4j + 4k$ Also the vector joining the point A(1, 1, – 1) and B(– 3, 5, 6) is given by $\overrightarrow{AB} = -4i + 4j + 7k$ $\therefore$ Unit vector along it is $\frac{-4i+4j+7k}{\sqrt{16+16+49}}$ i.e. $\frac{-4i+4j+7k}{9}$ $\therefore$ From equation (1), the directional derivative is $\frac{\partial\phi}{\partial r} = (8i-4j+4k) \cdot \left(\frac{-4i+4j+7k}{9}\right)$ $= \frac{-32-16+28}{9}$ $= -\frac{20}{9}$ $\therefore$ **The directional derivative** $= -\frac{20}{9}$ **Ans.**						
12	Unit 5	A particle moves along the curve Find the components of its velocity and acceleration at t = 1 in the direction of.	12	5	Med ium	Mode rate	CO-4	Numerical

		Soln. Given : $x = 2t^2$, $y = t^2 - 4t$, $z = 3t - 5$. If $\bar{r}$ is the position vector of any point P(x, y, z) on the given curve, then $\bar{r} = xi + yj + zk = 2t^2 i + (t^2 - 4t)j + (3t - 5)k$ Velocity, $\bar{v} = \frac{d\bar{r}}{dt} = 4ti + (2t - 4)j + 3k$ $= 4i - 2j + 3k$ at $t = 1$ Acceleration, $\bar{a} = \frac{d^2\bar{r}}{dt^2} = 4i + 2j$ $= 4i + 2j$ at $t = 1$ Now, the unit vector in the given direction $i - 3j + 2k$ $= \frac{i - 3j + 2k}{\sqrt{1 + 9 + 4}}$ $= \frac{i - 3j + 2k}{\sqrt{14}} = \hat{n}$ $\therefore$ The component of velocity in the given direction $= \bar{v}.\hat{n}$ $= (4i - 2j + 3k).\frac{i - 3j + 2k}{\sqrt{14}}$ $= \frac{4(1) - 2(-3) + 3(2)}{\sqrt{14}}$ $= \frac{16\sqrt{14}}{14}$ $= \frac{8\sqrt{14}}{7}$ The components of acceleration in the given direction $= \bar{a}.\hat{n}$ $= (4i + 2j).\frac{(i - 3j + 2k)}{\sqrt{14}}$ $= \frac{-2}{\sqrt{14}}$ $= \frac{-\sqrt{14}}{7}$ $\therefore$ The components of acceleration at $t = 1$ in the given direction $= -\frac{\sqrt{14}}{7}$. The components of velocity at $t = 1$ in the given direction $= \frac{8\sqrt{14}}{7}$. **Ans.**						
13	Unit 5	Find the directional derivative of at the point (2,-1,1) along the tangent to the curve at	12	5	Medium	Moderate	CO-4	Numerical

		Soln. Let $\phi = xy^2 + yz^3$ $\nabla\phi = \left(i\frac{\partial}{\partial x} + j\frac{\partial}{\partial y} + k\frac{\partial}{\partial z}\right)(xy^2 + yz^3)$ $= iy^2 + j(2xy + z^3) + k3z^2y$ $\therefore (\nabla\phi)_{(2,-1,1)} = i - 3j - 3k$ And $\bar{r} = xi + yj + zk = at\sin(t)\,i + a\cos(t)\,j + at\,k$ Tangent vector $\bar{T} = \frac{d\bar{r}}{dt} = (a\sin t + at\cos t)\,i - a\sin t\,j + ak$, $= a[(\sin t + t\cos t)\,i - \sin t\,j + k]$ $\therefore T|_{t=\frac{\pi}{4}} = a\left\{\left(\frac{1}{\sqrt{2}} + \frac{\pi}{4}\frac{1}{\sqrt{2}}\right) - \frac{1}{\sqrt{2}}j + k\right\}$ $T|_{t=\frac{\pi}{4}} = a\left\{\left(\frac{4+\pi}{4\sqrt{2}}\right)i - \frac{1}{\sqrt{2}}j + k\right\}$ $\therefore$ Required directional derivative $= (i - 3j - 3k) \cdot \frac{a\left\{\left(\frac{4+\pi}{4\sqrt{2}}\right)i - \frac{1}{\sqrt{2}}j + k\right\}}{a\sqrt{\frac{(4+\pi)^2}{32} + \frac{1}{2} + 1}}$ $= \frac{\frac{4+\pi}{4\sqrt{2}} + \frac{3}{\sqrt{2}} - 3}{\sqrt{\frac{16+\pi^2+8\pi+16+32}{32}}}$ $= \frac{4+\pi+12-12\sqrt{2}}{\sqrt{64+8\pi+\pi^2}}$ $= \frac{16+\pi-12\sqrt{2}}{\sqrt{64+8\pi+\pi^2}}$. $\therefore$ The required directional derivative $= \frac{16+\pi-12\sqrt{2}}{\sqrt{64+8\pi+\pi^2}}$ Ans.						
14	Unit 5	A particle moves along the curve	12	5	Med	Mode	CO-4	Numerical

		, where t is the time. Find the components of its velocity and acceleration at time in the direction 						
15	Unit 5	What is the directional derivative of $= x^2 - 2y^2 + 4z^2$ at the point (1, 1, -1) in the direction of . 	12	5	Medium	Mode rate	CO-4	Numerical
16	Unit 5	Find the rate of change of in the direction normal to the surface at	12	5	Med	Mode	CO-4	Numerical

		the point (1,1,1,).			ium	rate		
		Soln. We have $\phi = xyz$ $\Rightarrow \nabla\phi = \left(i\frac{\partial}{\partial x} + j\frac{\partial}{\partial y} + k\frac{\partial}{\partial z}\right)(xyz)$ $= iyz + jxz + kxy$ $\therefore (\nabla\phi)_{(1,1,1)} = i + j + k$						
		Normal to the surface $\psi = x^2y + y^2x + yz^2 - 3$ is given as $\nabla\psi = \left(i\frac{\partial}{\partial x} + j\frac{\partial}{\partial y} + k\frac{\partial}{\partial z}\right)(x^2y + y^2x + yz^2 - 3)$ $= i(2xy + y^2) + j(x^2 + 2xy + z^2) + k2yz$ $\therefore (\nabla\psi)_{(1,1,1)} = 3i + 4j + 2k$ $\therefore$ Unit normal vector $= \frac{3i+4j+2k}{\sqrt{9+16+4}} = \frac{3i+4j+2k}{\sqrt{29}}$ Required rate of change of $\phi = (i + j + k) \cdot \frac{(3i+4j+2k)}{\sqrt{29}} = \frac{3+4+2}{\sqrt{29}} = \frac{9}{\sqrt{29}}$. $\therefore$ The required rate of change of $\phi = \frac{9}{\sqrt{29}}$ Ans.						

17	Unit 5	Find the value of n for which the vector field will be solenoidal. Also check whether the vector field is irrotational.	12	5	Medium	Moderate	CO-4	Numerical

Soln. A vector $r^n \bar{r}$ is said to be solenoidal, if

$$\text{Div.}\,(r^n\bar{r}) = 0, \text{ where } \bar{r} = xi + yj + zk$$

$$\Rightarrow \nabla.(r^n\bar{r}) = 0$$

$$\Rightarrow \left(i\frac{\partial}{\partial x} + j\frac{\partial}{\partial y} + k\frac{\partial}{\partial z}\right).\left(r^n xi + r^n yj + r^n zk\right) = 0$$

$$\Rightarrow \frac{\partial}{\partial x}(r^n x) + \frac{\partial}{\partial y}(r^n y) + \frac{\partial}{\partial z}(r^n z) = 0$$

$$\Rightarrow r^n + xnr^{n-1}\frac{\partial r}{\partial x} + r^n + ynr^{n-1}\frac{\partial r}{\partial y} + r^n + nzr^{n-1}\frac{\partial r}{\partial z} = 0$$

$$\Rightarrow 3r^n + \left[x\frac{\partial r}{\partial x} + y\frac{\partial r}{\partial y} + z\frac{\partial r}{\partial z}\right] n r^{n-1} = 0$$

$$\Rightarrow 3r^n + \left(x \cdot \frac{x}{r} + y \cdot \frac{y}{r} + z \cdot \frac{z}{r}\right) r^{n-1} n = 0$$

$$\Rightarrow 3r^n + \frac{n}{r}(x^2 + y^2 + z^2)\, r^{n-1} = 0$$

$$\Rightarrow 3r^n + \frac{n}{r} r^2\, r^{n-1} = 0 \qquad \left[\because \bar{r} = xi + yj + zk,\ \therefore r^2 = x^2 + y^2 + z^2\right]$$

$$\Rightarrow 3r^n + nr^n = 0$$

$$\Rightarrow (3 + n)\, r^n = 0$$

$$\Rightarrow 3 + n = 0$$

$$\Rightarrow \boxed{n = -3}$$ **Ans.**

A vector $r^n\bar{r}$ is said to be irrotational, if $\nabla \times r^n\bar{r} = 0$

Consider

$$\nabla \times r^n\bar{r} = \begin{vmatrix} i & j & k \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ r^n x & r^n y & r^n z \end{vmatrix}$$

$$= i\left[\frac{\partial}{\partial y}(r^n z) - \frac{\partial}{\partial z} r^n y\right] + j\left[\frac{\partial}{\partial z}(r^n x) - \frac{\partial}{\partial x}(r^n z)\right] + k\left[\frac{\partial}{\partial x}(r^n y) - \frac{\partial}{\partial y}(r^n x)\right]$$

$$\Rightarrow \nabla \times r^n\bar{r} = \left[n\, r^{n-1}\, z \cdot \frac{\partial r}{\partial y} - nr^{n-1}\, y\frac{\partial r}{\partial z}\right] i + j\left[nr^{n-1}\, x\frac{\partial r}{\partial z} - nr^{n-1}\, z\frac{\partial r}{\partial x}\right] + k\left[nr^{n-1}\, y\frac{\partial r}{\partial x} - nr^{n-1}\, x\frac{\partial r}{\partial y}\right]$$

$$= \left[nr^{n-1}\, z \cdot \frac{y}{r} - nr^{n-1}\, y\frac{z}{r}\right] i + \left[nr^{n-1}\, x\frac{z}{r} - nr^{n-1}\, z\frac{x}{r}\right] j + \left[nr^{n-1}\, y\frac{x}{r} - nr^{n-1}\, x\frac{y}{r}\right] k$$

$= 0.$

$\therefore$ $r^n\bar{r}$ is irrotational.

$\therefore$ $\boxed{\text{The value of } n = -3 \text{ and } r^n\bar{r} \text{ is irrotational.}}$ **Ans.**

18	Unit 5	The temperature at any point in space is , determine the directional derivative of in the direction of the vector at the point (1,1,1).	12	5	Medium	Difficult	CO-4	Numerical

$T = xy + yz + zx$

$\frac{\partial T}{\partial x} = y + z \quad \frac{\partial T}{\partial y} = x + z \quad \frac{\partial T}{\partial z} = y + x$

$\therefore \nabla T = (y+z)\hat{i} + (x+z)\hat{j} + (y+x)\hat{k}$

$(\nabla T)_{(1,1,1)} = 2\hat{i} + 2\hat{j} + 2\hat{k} = 2(\hat{i} + \hat{j} + \hat{k})$

$\vec{p} = 3\hat{i} - 4\hat{k}$

$\hat{n} = \frac{\vec{p}}{|\vec{p}|} = \frac{3\hat{i} - 4\hat{k}}{\sqrt{9+16}} = \frac{1}{5}(3\hat{i} - 4\hat{k})$

$\therefore$ directional derivative $= \nabla T \cdot \hat{n}$

$= 2(\hat{i} + \hat{j} + \hat{k}) \cdot \frac{1}{5}(3\hat{i} - 4\hat{k})$

$= \frac{2}{5}(3 - 4)$

$= -\frac{2}{5}$.

19	Unit 5	Find the directional derivative of , where at the point (2, 0, 3) in the direction of the outward normal to the sphere at the point (3,2,1).	12	5	Medium	Difficult	CO-4	Numerical

Detailed Solution

Given $\bar{v} = xy^2\hat{i} + zy^2\hat{j} + xz^2\hat{k}$

$\therefore v^2 = \bar{v}\cdot\bar{v} = x^2y^4 + z^2y^4 + x^2z^4$

$\therefore \nabla v^2 = \frac{\partial v^2}{\partial x}\hat{i} + \frac{\partial v^2}{\partial y}\hat{j} + \frac{\partial v^2}{\partial z}\hat{k}$

$= (2xy^4 + 2xz^4)\hat{i} + (4x^2y^3 + 4z^2y^3)\hat{j}$
$+ (2zy^4 + 4x^2z^3)\hat{k}$

$\therefore [\nabla v^2]_{(2,0,3)} = 324\hat{i} + 0\hat{j} + 432\hat{k}$ — 2

Let $\phi = x^2 + y^2 + z^2 = 14$

$\nabla\phi = 2x\hat{i} + 2y\hat{j} + 2z\hat{k}$ $\therefore [\nabla\phi]_{(3,2,1)} = 6\hat{i} + 4\hat{j} + 2\hat{k}$

Let $\hat{n} = \frac{\nabla\phi}{|\nabla\phi|}$ be the unit normal vector — 1

$\hat{n} = \frac{6\hat{i} + 4\hat{j} + 2\hat{k}}{\sqrt{4x^2 + 4y^2 + 4z^2}} = \frac{6\hat{i} + 4\hat{j} + 2\hat{k}}{\sqrt{36 + 16 + 4}}$

$= \frac{3\hat{i} + 2\hat{j} + \hat{k}}{\sqrt{14}}$ — 1

$\therefore$ Directional derivative $= [\nabla v^2]_{(2,0,3)} \cdot \hat{n}$

$= (324\hat{i} + 0\hat{j} + 432\hat{k}) \cdot \frac{(3\hat{i} + 2\hat{j} + \hat{k})}{\sqrt{14}}$

$= \frac{972 + 0 + 432}{\sqrt{14}}$

$= \frac{1404}{\sqrt{14}}$ — 1

20	Unit 5	Show that (i)	12	5	Medium	Difficult	CO-4	Numerical	

Soln. (i) **To prove :** curl grad $\phi = 0$

We have $\nabla\phi = \frac{\partial\phi}{\partial x}i + \frac{\partial\phi}{\partial y}j + \frac{\partial\phi}{\partial z}k$

$$\therefore \text{curl grad } \phi = \nabla \times (\nabla\phi) = \begin{vmatrix} i & j & k \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ \frac{\partial\phi}{\partial x} & \frac{\partial\phi}{\partial y} & \frac{\partial\phi}{\partial z} \end{vmatrix}$$

$$= i\left[\frac{\partial^2\phi}{\partial y\partial z} - \frac{\partial^2\phi}{\partial y\partial z}\right] - j\left[\frac{\partial^2\phi}{\partial x\partial z} - \frac{\partial^2\phi}{\partial x\partial z}\right]$$

$$+ k\left[\frac{\partial^2\phi}{\partial x\partial y} - \frac{\partial^2\phi}{\partial x\partial y}\right]$$

$= 0\mathbf{i} + 0\mathbf{j} + 0\mathbf{k}$

$= 0$

$\therefore$ **curl grad $\phi = 0$** **Ans.**

(ii) **To prove :** div curl $\overline{A} = 0$

Let $\overline{A} = A_1\mathbf{i} + A_2\mathbf{j} + A_3\mathbf{k}$

Consider,

$$\overline{F} = \nabla \times \overline{A} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ A_1 & A_2 & A_3 \end{vmatrix}$$

$$= \mathbf{i}\left[\frac{\partial A_3}{\partial y} - \frac{\partial A_2}{\partial z}\right] + \mathbf{j}\left[\frac{\partial A_1}{\partial z} - \frac{\partial A_3}{\partial x}\right] + \mathbf{k}\left[\frac{\partial A_2}{\partial x} - \frac{\partial A_1}{\partial y}\right]$$

Now, let $\overline{F} = F_1\mathbf{i} + F_2\mathbf{j} + F_3\mathbf{k}$

$\therefore \quad F_1 = \left(\frac{\partial A_3}{\partial y} - \frac{\partial A_2}{\partial z}\right)$

$F_2 = \left(\frac{\partial A_1}{\partial z} - \frac{\partial A_3}{\partial x}\right)$ and

$F_3 = \left(\frac{\partial A_2}{\partial x} - \frac{\partial A_1}{\partial y}\right)$

$\therefore \quad \nabla.\overline{F} = \frac{\partial F_1}{\partial x} + \frac{\partial F_2}{\partial y} + \frac{\partial F_3}{\partial z}$

$$= \frac{\partial}{\partial x}\left(\frac{\partial A_3}{\partial y} - \frac{\partial A_2}{\partial z}\right) + \frac{\partial}{\partial y}\left(\frac{\partial A_1}{\partial z} - \frac{\partial A_3}{\partial x}\right) + \frac{\partial}{\partial z}\left(\frac{\partial A_2}{\partial x} - \frac{\partial A_1}{\partial y}\right)$$

$$= \frac{\partial^2 A_3}{\partial x \partial y} - \frac{\partial^2 A_2}{\partial x \partial z} + \frac{\partial^2 A_1}{\partial y \partial z} - \frac{\partial^2 A_3}{\partial x \partial y} + \frac{\partial^2 A_2}{\partial x \partial z} - \frac{\partial^2 A_1}{\partial y \partial z}$$

$= 0$

$\therefore$ **div curl $\overline{A} = 0$** **Ans.**

21	Unit 5	Find the directional derivative of the function at the pt. (2,-1,1) in the direction of the normal to the surface at (-1,2,1).	12	5	Medium	Difficult	CO-4	Numerical

$\phi = xy^2 + yz^3$

$\frac{\partial \phi}{\partial x} = y^2 \quad \frac{\partial \phi}{\partial y} = 2xy + z^3 \quad \frac{\partial \phi}{\partial z} = 3yz^2$

$\nabla \phi = y^2 i + (2xy + z^3) j + 3yz^2 k$

$(\nabla \phi)_{(2,-1,1)} = i - 3j - 3k$

$\psi = x \log z - y^2 + 4$

$\nabla \psi = \log z \, i - 2y j + \frac{x}{z} k$

$(\nabla \psi)_{(-1,2,1)} = 0i - 4j - k = -4j - k$

$\hat{n} = \frac{\nabla \psi}{|\nabla \psi|} = \frac{-4j - k}{\sqrt{16+1}} = \frac{-1}{\sqrt{17}} (4j + k)$

$\therefore$ Directional derivative $= \nabla \phi \cdot \hat{n}$

$= (i - 3j - 3k) \cdot \left(\frac{-1}{\sqrt{17}}\right)(4j + k)$

$= -\frac{1}{\sqrt{17}} (0 - 12 - 3)$

22	Unit 5	Find the directional derivative of the function at the pt. (1,1,-1) in the direction of . In what direction the directional derivative will be maximum and what is its magnitude. **Soln. Given :** $\phi = x^2 - 2y^2 + 4z^2$ $\therefore \nabla\phi = \left(i\frac{\partial}{\partial x} + j\frac{\partial}{\partial y} + k\frac{\partial}{\partial z}\right)(x^2 - 2y^2 + 4z^2)$ $= i(2x) + j(-4y) + k8z$ $\therefore [\nabla\phi]_{(1,1,-1)} = i2(1) + j(-4\times1) + k(8\times-1)$ $= 2i - 4j - 8k$ Let $\bar{a} = 2i + j - k$ $\therefore$ Unit vector in the direction of $\bar{a}$ is $\hat{a} = \frac{2i+j-k}{\sqrt{4+1+1}} = \frac{1}{\sqrt{6}}(2i + j - k)$ $\therefore$ Directional derivative of $\phi = \nabla\phi.\hat{a}$ $= (2i - 4j - 8k).\frac{1}{\sqrt{6}}(2i + j - k)$ $= \frac{1}{\sqrt{6}}(4 - 4 + 8)$ $= \frac{8}{\sqrt{6}}$ $\therefore$ The directional derivative of $\phi = \frac{8}{\sqrt{6}}$ Directional derivative is maximum in the direction of $\nabla\phi$ i.e $2i - 4j - 8k$ Its magnitude = $\lvert\nabla\phi\rvert = \sqrt{4+16+64} = \sqrt{84} = 2\sqrt{21}$ **Ans.**	12	5	Med ium	Diffi cult	CO-4	Numerical
23	Unit 5	Find the constants a, b, c so that is irrotational and hence find the function such that	12	5	Med ium	Diffi cult	CO-4	Numerical

Soln. We have, $\bar{F} = (x + 2y + az)i + (bx - 3y - z)j + (4x + cy + 2z)k$

$$\therefore \nabla \times \bar{F} = \begin{vmatrix} i & j & k \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ (x+2y+az) & (bx-3y-z) & (4x+cy+2z) \end{vmatrix}$$

$$= (c+1)i - (4-a)j + (b-2)k$$

As $\bar{F}$ is irrotational, $\nabla \times \bar{F} = \bar{0}$

$\Rightarrow (c+1)\,i - (4-a)\,j + (b-2)\,k = 0i + 0j + 0k$

$\therefore \quad c + 1 = 0,\ 4 - a = 0$ and $b - 2 = 0$

$\Rightarrow a = 4,\ b = 2,\ c = -1$

This gives

$\overline{F} = (x + 2y + 4z)\,i + (2x - 3y - z)\,j + (4x - y + 2z)\,k$

Now, we have to find ϕ such that $\overline{F} = \nabla\phi$

i.e. $(x + 2y + 4z)\,i + (2x - 3y - z)\,j + (4x - y + 2z)\,k$

$$= \frac{\partial\phi}{\partial x}\,i + \frac{\partial\phi}{\partial y}\,j + \frac{\partial\phi}{\partial z}\,k$$

$$\therefore \quad \frac{\partial\phi}{\partial x} = (x + 2y + 4z) \qquad \ldots\ldots (1)$$

$$\frac{\partial\phi}{\partial y} = (2x - 3y - z) \qquad \ldots\ldots (2)$$

$$\frac{\partial\phi}{\partial z} = (4x - y + 2z) \qquad \ldots\ldots (3)$$

Integrating eq. (1), eq. (2) and eq. (3) partially w.r. to x, y, z respectively, we get

$$\phi = \frac{x^2}{2} + 2yx + 4zx + \phi_1(y, z) \qquad \ldots\ldots (4)$$

$$\phi = 2xy - \frac{3y^2}{2} - zy + \phi_2(x, z) \qquad \ldots\ldots (5)$$

$$\phi = 4xz - yz + z^2 + \phi_3(x, y) \qquad \ldots\ldots (6)$$

Comparing eq. (4), eq. (5) and eq. (6), we get

$\phi_1(y, z)$ = terms in ϕ independent of x

$$= \frac{-3y^2}{2} - zy + z^2$$

$\phi_2(x, z)$ = terms in ϕ independent of y

$$= \frac{x^2}{2} + 4zx + z^2$$

$\phi_3(x, y)$ = terms in ϕ independent of z

$$= \frac{x^2}{2} + 2yx - 3y^2/2$$

Hence,

$$\phi = \frac{x^2}{2} + 2yx + 4zx - \frac{3y^2}{2} - zy + z^2$$

$$= \frac{x^2}{2} - \frac{3y^2}{2} + z^2 + 2xy + 4zx - zy$$

$\therefore$ $a = 4,\ b = 2,\ c = -1$ and

Scalar potential, $\phi = \frac{x^2}{2} - \frac{3y^2}{2} + z^2 + 2xy + 4zx - zy$

24	Unit 5	A vector field is given by Show that the field is irrotational and find the scalar potential. **Soln.** We have, $\bar{F} = (y^2 \cos x + z^3)\,i + (2y \sin x - 4)\,j + (3xz^2 + 2)\,k$ $\therefore \nabla \times \bar{F} = \begin{vmatrix} i & j & k \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ y^2 \cos x + z^3 & 2y \sin x - 4 & 3xz^2 + 2 \end{vmatrix}$ $= i(0-0) - j(3z^2 - 3z^2) + k(2y \cos x - 2y \cos x)$ $= 0$ $\therefore \nabla \times \bar{F} = 0$ $\therefore$ $\bar{F}$ is conservative **Ans.** Now, $\bar{F} = \nabla \phi$ $\Rightarrow (y^2 \cos x + z^3)\,i + (2y \sin x - 4)\,j + (3xz^2 + 2)\,k$ $= \frac{\partial \phi}{\partial x} i + \frac{\partial \phi}{\partial y} j + \frac{\partial \phi}{\partial z} k$ $\frac{\partial \phi}{\partial x} = y^2 \cos x + z^3$ ...(1) $\frac{\partial \phi}{\partial y} = 2y \sin x - 4$ ....(2) $\frac{\partial \phi}{\partial z} = 3xz^2 + 2$ ....(3) Integrating eq. (1), eq. (2) & eq. (3) partially w.r. to x, y, z, we get $\phi = y^2 \sin x + z^3 x + \phi_1(y, z)$ .....(4) $\phi = y^2 \sin x - 4y + \phi_2(x, z)$ .....(5) $\phi = xz^3 + 2z + \phi_3(x, y)$ .....(6) Comparing eq. (4), eq. (5) and eq. (6), we get $\phi_1(y, z)$ = terms in ϕ independent of $x = -4y + 2z$ $\phi_2(x, z)$ = terms in ϕ independent of $y = xz^3 + 2z$ $\phi_3(x, y)$ = terms in ϕ independent of $z = y^2 \sin x - 4y$ $\phi = y^2 \sin x + z^3 x - 4y + 2z$	12	5	Med ium	Diffi cult	CO-4	Numerical
25	Unit 5	Show that is a conservative vector field and find a function such that	12	5	Med	Diffi	CO-4	Numerical

		. Also find the work done in moving the particle from (1, -2, 1) to (3, 1, 4).			ium	cult		

Soln. Given : $\bar{F} = (2xy + z^3)\,i + x^2\,j + 3xz^2\,k$

$$\Rightarrow \nabla \times \bar{F} = \begin{vmatrix} i & j & k \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 2xy + z^3 & x^2 & 3xz^2 \end{vmatrix}$$

$$\therefore \quad \nabla \times \bar{F} = i\,(0) - j\,(3z^2 - 3z^2) + k\,(2x - 2x) = 0$$

Hence $\bar{F}$ is conservative **Ans.**

$$\therefore \quad \bar{F} = \nabla \phi$$

$$\therefore \quad (2xy + z^3)\,i + x^2\,j + 3xz^2\,k = i\frac{\partial \phi}{\partial x} + j\frac{\partial \phi}{\partial y} + k\frac{\partial \phi}{\partial z}$$

$$\therefore \quad \frac{\partial \phi}{\partial x} = 2xy + z^3 \qquad \ldots.(1)$$

$$\frac{\partial \phi}{\partial y} = x^2 \qquad \ldots.(2)$$

$$\frac{\partial \phi}{\partial z} = 3xz^2 \qquad \ldots.(3)$$

Integrating equations (1), (2) and (3) w. r. to x, y & z respectively, we get

$$\phi = x^2 y + z^3 x + \phi_1(y, z) \qquad \ldots(4)$$

$$\phi = x^2 y + \phi_2(x, z) \qquad \ldots(5)$$

$$\phi = x z^3 + \phi_3(x, y) \qquad \ldots(6)$$

Comparing equations (4), (5) and (6), we get

$\phi_1(y, z)$ = terms in ϕ independent of $x = 0$

$\phi_2(x, z)$ = terms in ϕ independent of $y = z^3 x$

		$\phi_3(x, y)$ = terms in ϕ independent of $z = x^2y$ Hence, $\phi = x^2y + z^3x$ $\therefore$ Required work done = $\phi_{(3,1,4)} - \phi_{(1,-2,1)}$ $= (9 + 192) - (-2 + 1)$ $= 202$ units $\therefore$ Scalar potential, $\phi = x^2y + z^3x$ and work done = 202 units **Ans.**						
26	Unit 5	If prove that (i) grad $= \frac{\vec{r}}{r}$ ∇ $\vec{r}$ (ii) grad(1/) = $-\frac{\vec{r}}{r^3}$ (iii) $r^n = nr^{n-2}$	15	5	Hig h	Very diffic ult	CO-4	Numerical

$\therefore\ r = \sqrt{x^2 + y^2 + z^2}$ Here $\bar{r} = xi + yj + zk$

$\Rightarrow\ r^2 = x^2 + y^2 + z^2$

$\therefore\ 2r\dfrac{\partial r}{\partial x} = 2x$ or

$\dfrac{\partial r}{\partial x} = \dfrac{x}{r}$

Similarly,

$\dfrac{\partial r}{\partial y} = \dfrac{y}{r}$ and $\dfrac{\partial r}{\partial z} = \dfrac{z}{r}$

Now,

$$\text{grad } r = \nabla r = \left(i\frac{\partial}{\partial x} + j\frac{\partial}{\partial y} + k\frac{\partial}{\partial z}\right) r$$

$$= i\frac{\partial r}{\partial x} + j\frac{\partial r}{\partial y} + k\frac{\partial r}{\partial z}$$

$$\therefore \text{ grad } r = i\frac{x}{r} + j\frac{y}{r} + k\frac{z}{r}$$

$$= \frac{xi + yj + zk}{r}$$

$$= \frac{\bar{r}}{r}$$

$$\therefore \quad \boxed{\text{grad } r = \frac{\bar{r}}{r}} \quad \textbf{Ans.}$$

(ii) $\text{grad}\left(\dfrac{1}{r}\right) = \nabla\left(\dfrac{1}{r}\right)$

$$= \left(i\frac{\partial}{\partial x} + j\frac{\partial}{\partial y} + k\frac{\partial}{\partial z}\right)\left(\frac{1}{r}\right)$$

$$= i\frac{\partial}{\partial x}\left(\frac{1}{r}\right) + j\frac{\partial}{\partial y}\left(\frac{1}{r}\right) + k\frac{\partial}{\partial z}\left(\frac{1}{r}\right)$$

$$= i\left(\frac{-1}{r^2}\frac{\partial r}{\partial x}\right) + j\left(\frac{-1}{r^2}\frac{\partial r}{\partial y}\right) + k\left(\frac{-1}{r^2}\frac{\partial r}{\partial z}\right)$$

$$= i\left(\frac{-1}{r^2}\frac{x}{r}\right) + j\left(\frac{-1}{r^2}\frac{y}{r}\right) + k\left(\frac{-1}{r^2}\frac{z}{r}\right)$$

$$\therefore \text{ grad}\left(\frac{1}{r}\right) = -\frac{x\,i + y\,j + z\,k}{r^3}$$

$$= -\frac{\bar{r}}{r^3}.$$

$$\therefore \quad \boxed{\text{grad}\left(\frac{1}{r}\right) = \frac{-\bar{r}}{r^3}} \quad \textbf{Ans.}$$

(iii) $$\nabla r^n = \left(\hat{i}\frac{\partial}{\partial x} + \hat{j}\frac{\partial}{\partial y} + \hat{k}\frac{\partial}{\partial z}\right)\left(\sqrt{x^2+y^2+z^2}\right)^n$$

$$\left\{\because\ r = |\bar{r}| = \sqrt{x^2+y^2+z^2}\right.$$

$$= \hat{i}\frac{\partial}{\partial x}\left(x^2+y^2+z^2\right)^{n/2} + \hat{j}\frac{\partial}{\partial y}\left(x^2+y^2+z^2\right)^{n/2}$$

$$+ \hat{k}\frac{\partial}{\partial z}\left(x^2+y^2+z^2\right)^{n/2}$$

$$= \hat{i}\frac{n}{2}\left(x^2+y^2+z^2\right)^{\frac{n}{2}-1} 2x + \hat{j}\frac{n}{2}\left(x^2+y^2+z^2\right)^{\frac{n}{2}-1} 2y$$

$$+ \hat{k}\frac{n}{2}\left(x^2+y^2+z^2\right)^{\frac{n}{2}-1} 2z$$

$$= n\left(x^2+y^2+z^2\right)^{\frac{n}{2}-1}\left[x\hat{i}+y\hat{j}+z\hat{k}\right]$$

$$= n\left(r^2\right)^{\frac{n-2}{2}}\bar{r}$$

$$= n\ r^{n-2}\bar{r}$$

27	Unit 5	If is a vector function and is a scalar function, then show that 3) $\text{Div}\ (\phi \bar{F}) = \nabla . (\phi \bar{F})$ $= \left(\hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z} \right) . (\phi \bar{F})$ $= \hat{i} . \frac{\partial}{\partial x} (\phi \bar{F}) + \hat{j} . \frac{\partial}{\partial y} (\phi \bar{F}) + \hat{k} . \frac{\partial}{\partial z} (\phi \bar{F})$ $\dot{=} \Sigma\, i . \frac{\partial}{\partial x} (\phi \bar{F})$ $= \Sigma\, i . \left(\phi \frac{\partial \bar{F}}{\partial x} + \frac{\partial \phi}{\partial x} \bar{F} \right)$ $= \Sigma \left\{ \hat{i} . \left(\phi \frac{\partial \bar{F}}{\partial x} \right) \right\} + \Sigma \left\{ \hat{i} . \left(\frac{\partial \phi}{\partial x} \bar{F} \right) \right\}$ $= \Sigma \left\{ \varphi \left(\hat{i} . \frac{\partial \bar{F}}{\partial x} \right) \right\} + \Sigma \left(\frac{\partial \varphi}{\partial x} \hat{i} \right) . \bar{F}, \quad \therefore\ \bar{a} . (m\bar{b}) = (ma) . \bar{b} = m (\bar{a} . \bar{b})$ $= \phi\, \Sigma\, i . \frac{\partial \bar{F}}{\partial x} + \Sigma \left(\frac{\partial \phi}{\partial x} \hat{i} \right) . \bar{F}$ $= \phi\, (\nabla . \bar{F}) + (\nabla \phi) . \bar{F}$ $= \phi\, div\, \bar{F} + (grad\ \phi) . \bar{F}$	15	5	High	Very difficult	CO-4	Numerical

28	Unit 5	Find the constant m and n such that the surface will be orthogonal to the surface at the point (1,-1,2). **Soln.** Let $\phi_1 = mx^2 - 2nyz - (m+4)x$ and $\phi_2 = 4x^2y + z^3 - 4$ The point P(1, –1, 2) lies on both surfaces. As this point lies in $mx^2 - 2nyz = (m+4)x$, we have $m - 2n(-2) = (m+4)$ $\Rightarrow m + 4n = m + 4$ $\Rightarrow n = 1$ $\therefore \phi_1 = mx^2 - 2yz - (m+4)x$ and $\phi_2 = 4x^2y + z^3 - 4$ Since these surfaces are orthogonal at (1, –1, 2), their normal at this point must be perpendicular. $\therefore$ [Vector normal to ϕ_1 at (1, –1, 2)] . [Vector normal to ϕ_2 at (1, –1, 2)] $= 0$ $\Rightarrow (\text{grad}\ \phi_1)_{\text{at }(1,-1,2)} \cdot (\text{grad}\ \phi_2)_{\text{at }(1,-1,2)} = 0$ ..... (1) Now, grad $\phi_1 = \nabla\phi_1$ $= \left(i\frac{\partial}{\partial x} + j\frac{\partial}{\partial y} + k\frac{\partial}{\partial z}\right).(mx^2 - 2yz - (m+4)x)$ $= [2mx - (m+4)]\,i + [-2z]\,j + [-2y]\,k$ $\therefore (\text{grad}\ \phi_1)_{\text{at}(1,-1,2)} = (m-4)\,i - 4j + 2k$ Also, grad $\phi_2 = \nabla\phi_2$ $= \left(i\frac{\partial}{\partial x} + j\frac{\partial}{\partial y} + k\frac{\partial}{\partial z}\right)(4x^2y + z^3 - 4)$ $= [8xy]\,i + [4x^2]\,j + [3z^2]\,k$ $\therefore (\text{grad}\ \phi_2)_{\text{at}(1,-1,2)} = -8\,i + 4j + 12\,k$ $\therefore$ Eq. (1) $\Rightarrow [(m-4)\,i - 4j + 2k] \cdot [-8i + 4j + 12k] = 0$ $\therefore -8(m-4) - 16 + 24 = 0$ $\Rightarrow -8m + 32 - 16 + 24 = 0$ $\Rightarrow -8m + 40 = 0$ $\therefore m = 5$ $\therefore \boxed{m = 5 \text{ and } n = 1}$ **Ans.**	15	5	High	Very difficult	CO-4	Numerical

29	Unit 5	Show that is irrotational. Find φ such that where **Soln.** We have, $\overline{E} = \frac{\bar{r}}{r^2}$ $\text{curl}\left(\frac{\bar{r}}{r^2}\right) = \frac{1}{r^2}\,\text{curl}\,\bar{r} + \left(\text{grad}\frac{1}{r^2}\right)\times\bar{r}$ $\left[\because \nabla\times(\varphi\overline{A}) = \varphi(\nabla\times\overline{A}) + (\nabla\varphi)\times\overline{A}\right]$ $= 0 + [-2r^{-4}\,\bar{r}]\times\bar{r}$ $[\because \text{curl}\,\bar{r} = 0 \text{ and grad } r^n = n r^{n-2}\bar{r}]$ $\therefore \text{curl}\,\overline{E} = \left(-2r^{-4}(\bar{r}\times\bar{r})\right) = 0$ $\therefore$ $\overline{E}$ is irrotational vector. **Ans.** To find scalar ϕ such that $\overline{E} = -\nabla\phi$ $\Rightarrow \frac{\bar{r}}{r^2} = -\left\{\frac{\partial\phi}{\partial x}i + \frac{\partial\phi}{\partial y}j + \frac{\partial\phi}{\partial z}k\right\}$ $\Rightarrow \frac{xi + yj + zk}{x^2+y^2+z^2} = -\frac{\partial\phi}{\partial x}i - \frac{\partial\phi}{\partial y}j - \frac{\partial\phi}{\partial z}k$ Equating the coefficients of i, j, k, we get $\frac{\partial\phi}{\partial x} = \frac{-x}{x^2+y^2+z^2}$ ..... (1) $\frac{\partial\phi}{\partial y} = \frac{-y}{x^2+y^2+z^2}$ ..... (2) $\frac{\partial\phi}{\partial z} = \frac{-z}{x^2+y^2+z^2}$ ..... (3) Integrating eqn. (1) partially w.r.to 'x', we get $\phi = \frac{-1}{2}\log(x^2+y^2+z^2) + \psi_1(y,z)$ ..... (4) Integrating eqn. (2) partially w.r.t. 'y', we have $\phi = \frac{-1}{2}\log(x^2+y^2+z^2) + \psi_2(z,x)$ ..... (5)	15	5	High	Very difficult	CO-4	Numerical

		Integrating eqn. (3) partially w.r.t. 'z', we have $\phi = \frac{-1}{2}\log(x^2+y^2+z^2) + \psi_3(x,y)$ Comparing eqn. (4), eqn. (5) and eqn. (6), we get $\psi_1(y,z) = \psi_2(z,x) = \psi_3(x,y) = 0$ Hence, $\phi = -\frac{1}{2}\log(x^2+y^2+z^2)$ $\therefore$ $\boxed{\phi = -\frac{1}{2}\log(x^2+y^2+z^2)}$ Ans.						
30	Unit 5	Prove that is conservative and find the scalar potential such that that .	15	5	Hig h	Very diffic ult	CO-4	Numerical

Soln. Since $\overline{F} = r^2\overline{r}$ and we have to prove that $\overline{F}$ is irrotational. (i.e. conservative).

$\therefore\ \overline{F} = r^2\overline{r} = r^2(xi + yj + zk)$

$\therefore\ \overline{F} = r^2xi + r^2yj + r^2zk$ and

$\overline{r} = xi + yj + zk$

$\Rightarrow r^2 = x^2 + y^2 + z^2$

$\Rightarrow 2r\dfrac{\partial r}{\partial x} = 2x$

$\therefore\ \dfrac{\partial r}{\partial x} = \dfrac{x}{r}$

Similarly, $\dfrac{\partial r}{\partial y} = \dfrac{y}{r}$ and $\dfrac{\partial r}{\partial z} = \dfrac{z}{r}$

$$\therefore \nabla \times \overline{F} = \begin{vmatrix} i & j & k \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ xr^2 & yr^2 & zr^2 \end{vmatrix}$$

$$= i\left(\frac{\partial}{\partial y}(zr^2) - \frac{\partial}{\partial z}(yr^2)\right) - j\left(\frac{\partial}{\partial x}(zr^2) - \frac{\partial}{\partial z}(xr^2)\right) + k\left(\frac{\partial}{\partial x}(yr^2) - \frac{\partial}{\partial y}(xr^2)\right)$$

$$= i\left[z2r\frac{\partial r}{\partial y} - y2r\frac{\partial r}{\partial z}\right] - j\left[z2r\frac{\partial r}{\partial x} - x2r\frac{\partial r}{\partial z}\right] + k\left[y2r\frac{\partial r}{\partial x} - x2r\frac{\partial r}{\partial y}\right]$$

$\therefore\ \nabla \times \overline{F} = i(2yz - 2yz) - j(2xz - 2xz) + k(2xy - 2xy) = 0$

$\therefore$ $\boxed{\overline{F} = r^2\overline{r} \text{ is irrotational. (i.e. conservative).}}$ **Ans.**

Now,

$\nabla\phi = \overline{F} = r^2(xi + yj + zk)$

$$\Rightarrow \frac{\partial\phi}{\partial x}i + \frac{\partial\phi}{\partial y}j + \frac{\partial\phi}{\partial z}k = xr^2 i + yr^2 j + zr^2 k$$

$$\frac{\partial\phi}{\partial x} = xr^2 = x(x^2 + y^2 + z^2) \quad \text{.....(1)}$$

$$\frac{\partial\phi}{\partial y} = y(x^2 + y^2 + z^2) \quad \text{....(2)}$$

$$\frac{\partial\phi}{\partial z} = z(x^2 + y^2 + z^2) \quad \text{.....(3)}$$

Integrating equations (1), (2), (3) w.r.to. x, y, z respectively, we get

$$\phi = \frac{x^4}{4} + \frac{x^2y^2}{2} + \frac{x^2z^2}{2} + \phi_1(y,z) \quad \text{.....(4)}$$

$$\phi = \frac{x^2y^2}{2} + \frac{y^4}{4} + \frac{y^2z^2}{2} + \phi_2(x,z) \quad \text{.....(5)}$$

$$\phi = \frac{x^2z^2}{2} + \frac{y^2z^2}{2} + \frac{z^4}{4} + \phi_3(x,y) \quad \text{.....(6)}$$

Comparing equations (4), (5) & (6), we get

$\phi_1(y,z)$ = terms in ϕ independent of x

$$= \frac{y^4}{4} + \frac{y^2z^2}{2} + \frac{z^4}{4}$$

$\phi_2(x,z)$ = terms in ϕ independent of y

$$= \frac{x^4}{4} + \frac{x^2z^2}{2} + \frac{z^4}{4}$$

$\phi_3(x,y)$ = terms in ϕ independent of z

$$= \frac{x^4}{4} + \frac{x^2y^2}{2} + \frac{y^4}{4}$$

Hence,

$$\phi = \frac{x^4}{4} + \frac{y^4}{4} + \frac{z^4}{4} + \frac{x^2y^2}{2} + \frac{y^2z^2}{2} + \frac{x^2z^2}{2}$$

$$\therefore \quad \boxed{\phi = \frac{x^4}{4} + \frac{y^4}{4} + \frac{z^4}{4} + \frac{x^2y^2}{2} + \frac{y^2z^2}{2} + \frac{x^2z^2}{2}} \quad \textbf{Ans.}$$

31	Unit 5	Show that $\bar{F} = (y^2 \cos x + z^3)i + (2y \sin x - 4)j + (3xz^2 + 2)k$ is a conservative vector field and find a function such that . Also find the work done in moving the particle from (0,1,-1) to (π/2, -1, 2). **Soln.** We have, $\bar{F} = (y^2 \cos x + z^3)i + (2y \sin x - 4)j + (3xz^2 + 2)k$ $\therefore \nabla \times \bar{F} = \begin{vmatrix} i & j & k \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ y^2 \cos x + z^3 & 2y \sin x - 4 & 3xz^2 + 2 \end{vmatrix}$ $= i(0 - 0) - j(3z^2 - 3z^2) + k(2y \cos x - 2y \cos x)$ $= 0$ $\therefore \nabla \times \bar{F} = 0$ $\therefore$ $\bar{F}$ is conservative **Ans.** Now, $\bar{F} = \nabla \phi$ $\Rightarrow (y^2 \cos x + z^3)i + (2y \sin x - 4)j + (3xz^2 + 2)k$ $= \frac{\partial \phi}{\partial x} i + \frac{\partial \phi}{\partial y} j + \frac{\partial \phi}{\partial z} k$ $\frac{\partial \phi}{\partial x} = y^2 \cos x + z^3$ ...(1) $\frac{\partial \phi}{\partial y} = 2y \sin x - 4$ ....(2) $\frac{\partial \phi}{\partial z} = 3xz^2 + 2$ ....(3) Integrating eq. (1), eq. (2) & eq. (3) partially w.r. to x, y, z , we get $\phi = y^2 \sin x + z^3 x + \phi_1(y, z)$ .....(4) $\phi = y^2 \sin x - 4y + \phi_2(x, z)$ .....(5) $\phi = xz^3 + 2z + \phi_3(x, y)$ .....(6) Comparing eq. (4), eq. (5) and eq. (6), we get $\phi_1(y, z)$ = terms in ϕ independent of $x = -4y + 2z$ $\phi_2(x, z)$ = terms in ϕ independent of $y = xz^3 + 2z$ $\phi_3(x, y)$ = terms in ϕ independent of $z = y^2 \sin x - 4y$ $\phi = y^2 \sin x + z^3 x - 4y + 2z$	15	5	High	Very difficult	CO-4	Numerical

Hence,

$\phi = y^2 \sin x + z^3 x - 4y + 2z$

Workdone - We know that in a conservative field, the work done depends on the value of ϕ at the end points A & B.

$\therefore$ Work done $= \phi(B) - \phi(A)$

$$= \phi\left(\frac{\pi}{2}, -1, 2\right) - \phi\,(0,\ 1, -1)$$

$$= (1 + 4\pi + 4 + 4) - (-4 - 2)$$

$$= 4\pi + 15.$$

$\therefore$ $\boxed{\phi = y^2 \sin x + z^3 x - 4y + 2z \text{ and Work done } = 4\pi + 15}$ **Ans.**